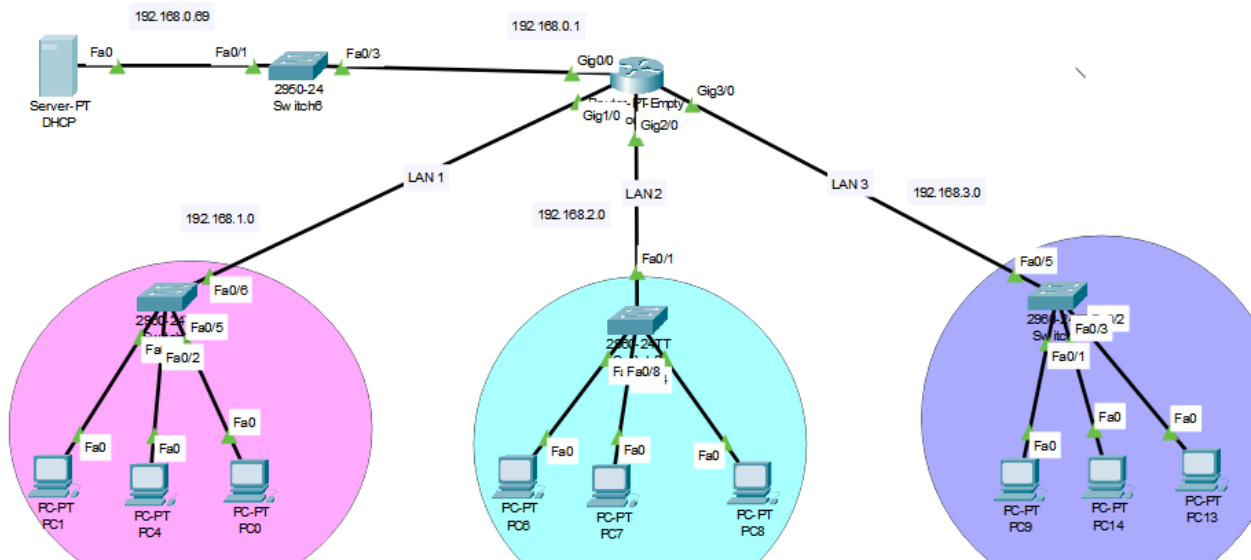


Lab Tasks

Task 01:

Design and configure a network where **one central DHCP server** dynamically assigns IP addresses to **three different LANs** through a router acting as a **DHCP Relay Agent**.

Network Requirements



Devices

- 1 Router
- 4 Switches
- 1 DHCP Server
- PCs for each subnet (as shown in the topology you will attach)

Addressing Information

Server Network

- Network: **192.168.xx.0/24**
- DHCP Server IP: **192.168.xx.69**
- Router Interface (Server side): **192.168.xx.1**

Client LAN Subnets

- LAN 1: **192.168.xx+1.0/24**
- LAN 2: **192.168.xx+2.0/24**
- LAN 3: **192.168.xx+3.0/24**
- [xx being the last two digits of your roll number]

TASK DESCRIPTION

1. Build the topology

Create the network using the devices listed above and connect links according to the given topology screenshot.

2. Configure router interfaces

Assign correct IPs to each router interface based on the addressing table above. Ensure all interfaces are enabled.

3. Configure DHCP Server

Create **three DHCP pools** one for each of the three client LANs

Each pool must include:

- Correct network address
- Correct subnet mask
- Correct default gateway
- A valid dynamic range

4. Verification and Testing

- Each PC receives an IP from the correct subnet
- Subnet mask and gateway values are correct
- Inter-subnet connectivity works
- DHCP Server shows active bindings
- No device receives an APIPA address (169.254.x.x)
- Pings between networks work

Task 02:

Your task is to configure IP NAT configuration in the below scenario and verify the results.

